

AthenaK post-processing computations and conventions

This document summarizes the computations and conventions needed to post-process a set of variables/quantities. It also collects some of the output variables that are given in the code and how they are defined.

Conventions

VARIABLES

- Metric: The spatial metric is defined with lower indices, i.e. γ_{ij} . It can be extracted from the code by specifying the output variables `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, and `adm_gzz` or by giving the block of variables `adm`, which also contains the extrinsic curvature K_{ij} . Note that the `<adm>` block needs to exist for this to apply.
- Lapse: As a scalar quantity, the lapse α follows no convention and can be fetched by specifying the `z4c_alpha` variable in the respective `<output>` block. Note that the `<z4c>` block needs to exist for this to apply.
- Shift: Together with the lapse α , the shift completes the set of gauge variables and is defined with upper indices, i.e. β^i . It can be specified in the output via `z4c_betax`, `z4c_betay`, and `z4c_betaz`. Note that also here the `<z4c>` block needs to exist.
- Magnetic field components: In AthenaK, we evolve the densitized magnetic field components, i.e. $\tilde{B}^i = \sqrt{\gamma} B^i$ with upper indices. By specifying the output for the magnetic field components via `bcc1`, `bcc2`, and `bcc3`, we therefore mean $\tilde{B}^i$ cell-centered on the grid. $\tilde{B}^i$ is defined in the Eulerian frame and needs to be undensitized in the post-processing step. This requires the metric (see above). Note that in this case the `<mhd>` block needs to exist.
- 3-velocity: The convention for the 3-velocity in AthenaK is a bit tricky. When we specify `velx`, `vely`, and `velz` in the output, then we mean the Lorentz-weighted 3-velocity in the fluid frame $\tilde{u}^i = W v^i$, where v^i is the 3-velocity in the Eulerian frame and W the Lorentz factor. In GR-Athena++ the variable $\tilde{u}^i$ gets directly evolved and is defined as the projection of the 4-velocity u^μ onto the hypersurface Σ_t . This is important to know when computing the 4-velocity in some post-processing steps below. For the output of the Lorentz-weighted 3-velocity either `<hydro>` or `<mhd>` needs to exist.
- M1 variables: The neutrino energy density E follows no convention as a scalar quantity and can be extracted via `rad_m1_E`. On the other hand, the neutrino number flux F_i is given with lower indices as `rad_m1_F`, and lastly the neutrino number density also follows no convention as `rad_m1_N`. All the aforementioned variables are given per species in the Eulerian frame. To obtain all of them in the fluid frame, one would have to project the stress-energy tensor accordingly.

DERIVED VARIABLES Here we collect a couple of derived variables, that are needed for the computation of some post-processing quantities below:

- Lorentz-factor: The Lorentz factor in terms of the AthenaK conventions can be computed via

$$W = \sqrt{1 + \tilde{u}^i \tilde{u}_i} = \sqrt{1 + \gamma_{ij} \tilde{u}^i \tilde{u}^j} \mid \tilde{u}^i = W v^i,$$

which is equal to the common form by realizing that

$$W = \sqrt{1 + \tilde{u}^i \tilde{u}_i} = \sqrt{1 + W^2 v^2} = \sqrt{1 + \frac{v^2}{1 - v^2}} = \sqrt{\frac{1 - v^2 + v^2}{1 - v^2}} = \sqrt{\frac{1}{1 - v^2}}.$$

Writing this in terms of the AthenaK variables gives

$$\begin{aligned} \gamma_{ij} \tilde{u}^i \tilde{u}^j &= \text{adm_gxx velx velx} + 2 \text{adm_gxy velx vely} + 2 \text{adm_gxz velx velz} \\ &\quad + \text{adm_gyy vely vely} + 2 \text{adm_gyz vely velz} + \text{adm_gzz velz velz}, \\ W &= \sqrt{1 + \gamma_{ij} \tilde{u}^i \tilde{u}^j}. \end{aligned} \tag{1}$$

- Determinant of the spatial metric: To get the determinant of the spatial metric γ_{ij} , we can first write it in component form as

$$\gamma = \gamma_{xx}\gamma_{yy}\gamma_{zz} - 2\gamma_{xy}\gamma_{xz}\gamma_{yz} - \gamma_{xx}\gamma_{yz}\gamma_{yz} - \gamma_{yy}\gamma_{xz}\gamma_{xz} - \gamma_{zz}\gamma_{xy}\gamma_{xy}.$$

In terms of the AthenaK variables this becomes

$$\gamma = \text{adm_gxx adm_gyy adm_gzz} - 2\text{adm_gxy adm_gxz adm_gyz} - \text{adm_gxx adm_gyz adm_gyz} - \text{adm_gyy adm_gxz adm_gxz} - \text{adm_gzz adm_gxy adm_gxy}. \quad (2)$$

- Inverse spatial metric components: Since AthenaK only provides γ_{ij} , we need to calculate the inverse spatial metric γ^{ij} with help of the determinant of the metric which we can obtain via Equ. (??). The components are then computed via

$$\begin{aligned} \gamma^{xx} &= (\gamma_{yy}\gamma_{zz} - \gamma_{yz}^2)/\gamma, \\ \gamma^{yy} &= (\gamma_{xx}\gamma_{zz} - \gamma_{xz}^2)/\gamma, \\ \gamma^{zz} &= (\gamma_{xx}\gamma_{yy} - \gamma_{xy}^2)/\gamma, \\ \gamma^{xy} &= (\gamma_{xz}\gamma_{yz} - \gamma_{xy}\gamma_{zz})/\gamma, \\ \gamma^{xz} &= (\gamma_{xy}\gamma_{yz} - \gamma_{xz}\gamma_{yy})/\gamma, \\ \gamma^{yz} &= (\gamma_{xy}\gamma_{xz} - \gamma_{yz}\gamma_{xx})/\gamma. \end{aligned}$$

Or in terms of AthenaK variables

$$\gamma^{xx} = (\text{adm_gyy adm_gzz} - \text{adm_gyz}^2)/\gamma, \quad (3)$$

$$\gamma^{yy} = (\text{adm_gxx adm_gzz} - \text{adm_gxz}^2)/\gamma, \quad (4)$$

$$\gamma^{zz} = (\text{adm_gxx adm_gyy} - \text{adm_gxy}^2)/\gamma, \quad (5)$$

$$\gamma^{xy} = (\text{adm_gxz adm_gyz} - \text{adm_gxy adm_gzz})/\gamma, \quad (6)$$

$$\gamma^{xz} = (\text{adm_gxy adm_gyz} - \text{adm_gxz adm_gyy})/\gamma, \quad (7)$$

$$\gamma^{yz} = (\text{adm_gxy adm_gxz} - \text{adm_gyz adm_gxx})/\gamma. \quad (8)$$

- Undensitized Eulerian magnetic field components: Since AthenaK stores the densitized magnetic field components $\tilde{B}^i$, we need to undensitize them using Equ. (??), i.e.

$$B^i = \frac{\tilde{B}^i}{\sqrt{\gamma}}.$$

Or in terms of AthenaK variables

$$B^x = \frac{\text{bcc1}}{\sqrt{\gamma}}, \quad (9)$$

$$B^y = \frac{\text{bcc2}}{\sqrt{\gamma}}, \quad (10)$$

$$B^z = \frac{\text{bcc3}}{\sqrt{\gamma}}. \quad (11)$$

- Magnetic field amplitude in the Eulerian frame: Using the metric components γ_{ij} and the undensitized magnetic field components of Eqs. (??)-(??), we can compute the magnetic field amplitude in the Eulerian frame via

$$B^2 = \gamma_{ij}B^iB^j.$$

Or in terms of the AthenaK variables

$$\begin{aligned} B^2 &= \text{adm_gxx bcc1 bcc1} + 2\text{adm_gxy bcc1 bcc2} + 2\text{adm_gxz bcc1 bcc3} \\ &+ \text{adm_gyy bcc2 bcc2} + 2\text{adm_gyz bcc2 bcc3} + \text{adm_gzz bcc3 bcc3}. \end{aligned} \quad (12)$$

- Magnetic field amplitude in the fluid frame: To get the magnetic field amplitude in the fluid frame, we require again the metric components γ_{ij} , the undensitized magnetic field components B^i , and the Lorentz-weighted 3-velocity in the Lorentz frame $\tilde{u}^i$. In general, the magnetic field amplitude in the fluid frame can be written as

$$b^2 = \frac{(B^2 + (B^i W v_i)(B^i W v_i))}{W^2}. \quad (13)$$

We can make use of Equ. (??) and Equ. (??). The remaining two terms can be computed as

$$\begin{aligned} B^i W v_i &= B^i \tilde{u}_i = B^i \gamma_{ij} \tilde{u}^j = \gamma_{xx} B^x \tilde{u}^x + \gamma_{yy} B^y \tilde{u}^y + \gamma_{zz} B^z \tilde{u}^z \\ &+ \gamma_{xy} (B^x \tilde{u}^y + B^y \tilde{u}^x) + \gamma_{xz} (B^x \tilde{u}^z + B^z \tilde{u}^x) + \gamma_{yz} (B^y \tilde{u}^z + B^z \tilde{u}^y). \end{aligned}$$

Or in terms of the AthenaK variables:

$$\begin{aligned} B^i \gamma_{ij} \tilde{u}^i &= \text{adm_gxx velx bcc1} + \text{adm_gyy bcc2 vely} + \text{adm_gzz bcc3 velz} \\ &+ \text{adm_gxy (bcc1 vely + bcc2 velx)} + \text{adm_gxz (bcc1 velz + bcc3 velx)} \\ &+ \text{adm_gyz (bcc2 velz + bcc3 vely)}. \end{aligned} \quad (14)$$

Combining Eqs. (??), (??), and (??) in the definition for b^2 , yields the desired result.

- 4-velocity with upper indices: To obtain the 4-velocity u^μ , we require the metric components γ_{ij} , the 3-velocity $\tilde{u}^i$, and the gauge variables α and β^i . The definitions are

$$\begin{aligned} u^t &= \frac{W}{\alpha}, \\ u^i &= \tilde{u}^i - u^t \beta^i. \end{aligned}$$

By using Eqs. (??), we can use the AthenaK variables to write

$$u^t = \frac{W}{\text{z4c_alpha}}, \quad (15)$$

$$u^x = \text{velx} - u^t \text{z4c_betax}, \quad (16)$$

$$u^y = \text{vely} - u^t \text{z4c_betay}, \quad (17)$$

$$u^z = \text{velz} - u^t \text{z4c_betaz}. \quad (18)$$

- 4-velocity with lower indices: To compute the 4-velocity with lower indices, we need the gauge variables α and β^i , as well as the metric components γ_{ij} and 3-velocity. We start by writing down the general lowering operation

$$\begin{aligned} u_t &= g_{t\nu} u^\nu = g_{tt} u^t + g_{ti} u^i = (-\alpha^2 + \beta^k \beta_k) u^t + \beta_i u^i = -\alpha W + \beta^i u_i, \\ u_i &= g_{i\nu} u^\nu = g_{it} u^t + g_{ij} u^j = \beta_i u^t + \gamma_{ij} u^j = \gamma_{ij} \beta^j u^t + \gamma_{ij} u^j = \gamma_{ij} (u^j + \beta^j u^t). \end{aligned}$$

To reach the first equality, we have contracted with β^i and realized from above that $u^t = \frac{W}{\alpha}$, i.e.

$$\begin{aligned} u_t &= -\alpha W + \beta^k \beta_k u^t + \beta_i u^i = -\alpha W + \beta^i (\beta_i u^t) + \beta^i (\gamma_{ij} u^j) \\ &= \beta^i (\beta_i u^t + \gamma_{ij} u^j) \\ &= \beta^i u_i. \end{aligned}$$

In the second equality, we have relabeled and reordered the shift components in the first term, while in the second we have raised the index of the shift vector and also reordered. The last equality uses the general expression for u_i above. We can also realize that the bracket in u_i , i.e. $u^j + \beta^j u^t$, is equal to the 3-velocity AthenaK uses, redefined in terms of the 4-velocity, see Eqs. (??)-(??). Therefore, we have $u^j + \beta^j u^t = \tilde{u}^j$. We can therefore express the 4-velocity with lowered indices in terms of the AthenaK variables as

$$u_x = \text{adm_gxx velx} + \text{adm_gxy vely} + \text{adm_gxz velz}, \quad (19)$$

$$u_y = \text{adm_gxy velx} + \text{adm_gyy vely} + \text{adm_gyz velz}, \quad (20)$$

$$u_z = \text{adm_gxz velx} + \text{adm_gyz vely} + \text{adm_gzz velz}, \quad (21)$$

$$u_t = -\text{z4c_alpha} W + \beta_x u_x + \beta_y u_y + \beta_z u_z. \quad (22)$$

The shift with the lowered components can be calculated via Eqs. (??)-(??).

- Magnetic field components in fluid frame with upper indices: To obtain the magnetic field components in the fluid frame with upper indices b^μ , we require the gauge variables α and β^i , as well as the undensitized magnetic field components in the Eulerian frame B^i , the Lorentz factor W , the 3-velocity $\tilde{u}^i$ and the spatial components of the 4-velocity u^μ

$$b^t = \frac{B^i W \tilde{u}_i}{\alpha},$$

$$b^i = \frac{B^i + \alpha b^t u^i}{W}.$$

After computing $B^i W \tilde{u}_i$ via Equ. (??), undensitizing the Eulerian magnetic field components with Eqs. (??)-(??), computing the Lorentz factor with Equ. (??), and computing the four velocity components with Eqs. (??)-(??), we can compute the magnetic field components in the fluid frame via

$$b^t = \frac{B^i W \tilde{u}_i}{\text{z4c_alpha}}, \quad (23)$$

$$b^x = \frac{\text{bcc1} + \text{z4c_alpha} b^t u^x}{W}, \quad (24)$$

$$b^y = \frac{\text{bcc2} + \text{z4c_alpha} b^t u^y}{W}, \quad (25)$$

$$b^z = \frac{\text{bcc3} + \text{z4c_alpha} b^t u^z}{W}. \quad (26)$$

- Shift with lower components: Since AthenaK only outputs β^i , it is instructive for some computations to compute the shift with lower components, i.e. β_i , via

$$\begin{aligned} \beta_i &= \gamma_{ik} \beta^k, \\ \beta_x &= \gamma_{xx} \beta^x + \gamma_{xy} \beta^y + \gamma_{xz} \beta^z, \\ \beta_y &= \gamma_{yx} \beta^x + \gamma_{yy} \beta^y + \gamma_{yz} \beta^z, \\ \beta_z &= \gamma_{zx} \beta^x + \gamma_{zy} \beta^y + \gamma_{zz} \beta^z. \end{aligned}$$

To achieve this, we require the metric components γ_{ij} from the output, as well as the shift with upper components, or in terms of AthenaK variables we can write

$$\beta_x = \text{adm_gxx z4c_betax} + \text{adm_gxy z4c_betay} + \text{adm_gxz z4c_betaz}, \quad (27)$$

$$\beta_y = \text{adm_gxy z4c_betax} + \text{adm_gyy z4c_betay} + \text{adm_gyz z4c_betaz}, \quad (28)$$

$$\beta_z = \text{adm_gxz z4c_betax} + \text{adm_gyz z4c_betay} + \text{adm_gzz z4c_betaz}. \quad (29)$$

- Amplitude of the shift vector: For the computation of the amplitude of the shift vector, we require the AthenaK shift with upper indices, i.e. β^i , as well as the metric components. It can then be computed via

$$\beta^2 = \gamma_{ij} \beta^i \beta^j = \gamma_{xx} \beta^x \beta^x + 2\gamma_{xy} \beta^x \beta^y + 2\gamma_{xz} \beta^x \beta^z + \gamma_{yy} \beta^y \beta^y + 2\gamma_{yz} \beta^y \beta^z + \gamma_{zz} \beta^z \beta^z.$$

Or in terms of AthenaK variables

$$\begin{aligned} \beta^2 &= \text{adm_gxx z4c_betax z4c_betax} + 2 \text{adm_gxy z4c_betax z4c_betay} + 2 \text{adm_gxz z4c_betax z4c_betaz} \\ &+ \text{adm_gyy z4c_betay z4c_betay} + 2 \text{adm_gyz z4c_betay z4c_betaz} + \text{adm_gzz z4c_betaz z4c_betaz}. \end{aligned} \quad (30)$$

- Magnetic field components in fluid frame with lower indices: To obtain the magnetic field components in the fluid frame with lower indices b_μ , we require the shift β^i , the magnetic field components in the fluid frame with upper indices b^μ , and the metric components γ_{ij} . The general lowering operation is given by

$$\begin{aligned} b_t &= g_{\mu t} b^\mu = g_{tt} b^t + g_{tj} b^j = (-\alpha^2 + \beta^2) b^t + \beta_j b^j, \\ b_j &= g_{\mu j} b^\mu = g_{tj} b^t + g_{ij} b^i = \beta_j b^t + \gamma_{ij} b^i. \end{aligned}$$

The latter equality is given by the line element in 3+1 notation

$$\begin{aligned} ds^2 &= -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt), \\ g_{tt} &= -\alpha^2 + \beta_i \beta^i = -\alpha^2 + \beta^2, \\ g_{ti} &= \beta_i, \\ g_{ij} &= \gamma_{ij}. \end{aligned}$$

After using Eqs. (??)-(??) to compute the magnetic field components in the fluid frame with upper indices, Equ. (??) to compute the amplitude of the shift vector and Eqs. (??)-(??) to compute the shift vector with lower indices, we can use the AthenaK variables to get the lower indices of the magnetic field components in the fluid frame via

$$b_t = (-\text{z4c_alpha}^2 + \beta^2)b^t + \beta_x b^x + \beta_y b^y + \beta_z b^z, \quad (31)$$

$$b_x = \text{adm_gxx}(b^t \text{z4c_betax} + b^x) + \text{adm_gxy}(b^t \text{z4c_betay} + b^y) + \text{adm_gxz}(b^t \text{z4c_betaz} + b^z), \quad (32)$$

$$b_y = \text{adm_gxy}(b^t \text{z4c_betax} + b^x) + \text{adm_gyy}(b^t \text{z4c_betay} + b^y) + \text{adm_gyz}(b^t \text{z4c_betaz} + b^z), \quad (33)$$

$$b_z = \text{adm_gxz}(b^t \text{z4c_betax} + b^x) + \text{adm_gyz}(b^t \text{z4c_betay} + b^y) + \text{adm_gzz}(b^t \text{z4c_betaz} + b^z). \quad (34)$$

- Local angular velocity: To compute the local angular velocity, we require the AthenaK 3-velocity $\tilde{u}^i$, the metric components γ_{ij} and the gauge variables. In general we have

$$\Omega = \frac{u^\phi}{u^t}. \quad (35)$$

The velocity component ϕ is defined in terms of the Jacobian and the spatial components of the 4-velocity, i.e.

$$u^\phi = \frac{\partial \phi}{\partial x^i} u^i.$$

With $\phi = \arctan\left(\frac{y}{x}\right)$, we can compute the local angular velocity on any plane. For instance, in the xy-plane we find

$$\begin{aligned} u^\phi &= \frac{\partial \phi}{\partial x} u^x + \frac{\partial \phi}{\partial y} u^y = -\frac{y}{x^2 + y^2} u^x + \frac{x}{x^2 + y^2} u^y = \frac{xu^y - yu^x}{R^2}, \\ \Omega &= \frac{xu^y - yu^x}{R^2 u^t}. \end{aligned}$$

Here, the 4-velocity components with upper indices can be computed via Eqs. (??)-(??).

- Radial component of a 3-vector: For the computation of the radial component of a 3-vector with components (p^x, p^y, p^z) , we can write

$$p^r = p^x \sin \theta \cos \phi + p^y \sin(\theta) \sin \phi + p^z \cos \theta. \quad (36)$$

- Raise spatial vector: To raise a spatial vector with lower indices (p_x, p_y, p_z) , we can use the inverse spatial metric components from Eqs. (??)-(??) to get

$$p^x = \gamma \quad x p^x + \gamma^{xy} p^y + \gamma^{xz} p^z, \quad (37)$$

$$p^y = \gamma \quad x y p^x + \gamma^{yy} p^y + \gamma^{yz} p^z, \quad (38)$$

$$p^z = \gamma \quad x z p^x + \gamma^{yz} p^y + \gamma^{zz} p^z. \quad (39)$$

- Metric-normalized radial projection: To project a vector p^i along $\frac{r_i}{\sqrt{g^{ij} r_i r_j}}$, we first compute the inverse spatial metric via Eqs. (??)-(??). The inverse 4-metric g^{ij} is then obtained via

$$\begin{aligned} g^{xx} &= \gamma^{xx} - (\beta^x \beta^x) / \alpha^2, \\ g^{yy} &= \gamma^{yy} - (\beta^y \beta^y) / \alpha^2, \\ g^{zz} &= \gamma^{zz} - (\beta^z \beta^z) / \alpha^2, \\ g^{xy} &= \gamma^{xy} - (\beta^x \beta^y) / \alpha^2, \\ g^{xz} &= \gamma^{xz} - (\beta^x \beta^z) / \alpha^2, \\ g^{yz} &= \gamma^{yz} - (\beta^y \beta^z) / \alpha^2. \end{aligned}$$

The components of the vector r_i are computed with the angles of the sphere

$$\begin{aligned} r_x &= r \sin \theta \cos \phi, \\ r_y &= r \sin \theta \sin \phi, \\ r_z &= r \cos \theta \end{aligned}$$

The denominator $\sqrt{g^{ij}r_i r_j}$ is determined as

$$\sqrt{g^{ij}r_i r_j} = \sqrt{g^{xx}r_x^2 + g^{yy}r_y^2 + g^{zz}r_z^2 + 2g^{xy}r_x r_y + 2g^{xz}r_x r_z + 2g^{yz}r_y r_z}.$$

Lastly the projection of the vector p^i is defined as

$$p_{\text{norm}}^r = \frac{r_x p^x + r_y p^y + r_z p^z}{\sqrt{g^{ij}r_i r_j}}. \quad (40)$$

Computations

POYNTING FLUX For the computation of the Poynting flux across a shell, we require a set of output variables interpolated onto a sphere of fixed radius and specified in the parameter file via:

```
<output1>
file_type = sph
variable  = velx
dt        = 10.0
radius    = 400.0
ntheta    = 256
```

The variables we require are:

- The ADM metric components `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, and `adm_gzz` or the entire ADM output block `adm`. Note: if the entire `adm` block is specified, then the six extrinsic curvature components are also stored which are not needed for this computation and might fill unnecessary disk space.
- The gauge variables `z4c_alpha`, `z4c_betax`, `z4c_betay`, and `z4c_betaz`.
- The Lorentz-weighted 3-velocity $\tilde{u}^i$, given by the output variables `velx`, `vely`, and `velz`.
- The densitized magnetic field components in the Eulerian frame $\tilde{B}^i$, given by `bcc1`, `bcc2`, and `bcc3`.

A step-by-step approach is:

1. Compute the Lorentz-factor via Equ. (??).
2. Compute the 4-velocity with upper indices u^i using Eqs. (??)-(??).
3. Compute the lower time-component of the 4-velocity u_t via Equ. (??).
4. Compute the determinant of the metric γ via (Equ. (??)), to undensitize the magnetic field components with Eqs. (??)-(??).
5. Compute B^2 in the Eulerian frame via Equ. (??) and $B^i W \tilde{u}_i$ via Equ. (??).
6. Compute b^2 in the fluid frame through Equ. (??).
7. Compute the magnetic field components in the fluid frame b^μ via Eqs. (??)-(??).
8. Lower the indices on the shift vector (β_i) via Eqs. (??)-(??).
9. Compute β^2 with Equ. (??).
10. Compute the lower time-component of the magnetic field in the fluid frame, i.e. b_t , with Equ. (??).

11. Compute the radial component of the velocity vector (u^x, u^y, u^z) and the magnetic field (b^x, b^y, b^z) (in the fluid frame) with Equ. (??) (u^r and b^r).
12. Compute the integrand as

$$I = (b^2 u^r u_t - b^r b_t) \alpha \sqrt{\gamma} r^2 \sin \theta.$$

The spherical coordinates (r, θ, ϕ) can be obtained from the spherical output `sph`.

13. Lastly, integrate I over the sphere with angles (θ, ϕ) as dictated by the spherical output.

■ **λ_{MRI} AND QUALITY FACTOR Q** For the computation of the λ_{MRI} parameter and the quality factor Q , we ideally require 3D binary output. However, this procedure also works on a 2D slice. In general, we require

```
<output2>
file_type = bin
variable = dens
dt = 10.0
slice_x3 = 0.0 # optional
```

The variables we require are:

- The density ρ given by `dens`, the temperature `temperature`, the electron fraction Y_e `s_00`, the densitized magnetic field components `bcc1`, `bcc2`, `bcc3`, and the Lorentz-weighted 3-velocity components $\tilde{u}^i$, given by `velx`, `vely`, and `velz`.
- The ADM metric components `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, and `adm_gzz` or the entire ADM output block `adm`. Note: if the entire `adm` block is specified, then the six extrinsic curvature components are also stored which are not needed for this computation and might fill unnecessary disk space.
- The gauge variables `z4c_alpha`, `z4c_betax`, `z4c_betay`, and `z4c_betaz`.

A step-by-step approach is:

1. Compute the Lorentz-factor via Equ. (??).
2. Compute the determinant of the metric γ via (Equ. (??)), to undensitize the magnetic field components with Eqs. (??)-(??).
3. Compute B^2 in the Eulerian frame via Equ. (??) and $B^i W \tilde{u}_i$ via Equ. (??).
4. Compute b^2 in the fluid frame through Equ. (??).
5. Compute the magnetic field components in the fluid frame b^μ via Eqs.(??)-(??).
6. Compute the 4-velocity with upper indices u^i using Eqs. (??)-(??).
7. Compute the spatial magnetic field components in the fluid frame with lower indices b_i via Eqs. (??)-(??).
8. Compute the local angular velocity either on the plane or if 3D over the entire domain via Equ. (??).
9. The components of λ_{MRI} are then computed via

$$\lambda_{\text{MRI},i} = \frac{2\pi b_i}{\Omega \sqrt{\rho h + b^2}},$$

where h is the enthalpy given by the equation of state and the state vector (n_B, Y_e, T) at a given point.

10. The quality average quality factor for each direction is then

$$Q_i = \langle \lambda_{\text{MRI},i} / \Delta x \rangle,$$

where Δx is the grid spacing at the given grid point.

EJECTA - BERNOULLI AND GEODESIC CRITERION To compute the ejecta properties based on the Bernoulli and Geodesic criterion, we need data interpolated onto a sphere at a fixed radius, i.e.

```
<output3>
file_type = sph
variable = velx
dt = 10.0
radius = 400.0
ntheta = 256
```

The variables we require are:

- The density ρ given by `dens`, the temperature `temperature`, the electron fraction Y_e `s_00`, and the Lorentz-weighted 3-velocity components $\tilde{u}^i$, given by `velx`, `vely`, and `velz`.
- The ADM metric components `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, and `adm_gzz` or the entire ADM output block `adm`. Note: if the entire `adm` block is specified, then the six extrinsic curvature components are also stored which are not needed for this computation and might fill unnecessary disk space.
- The gauge variables `z4c_alpha`, `z4c_betax`, `z4c_betay`, and `z4c_betaz`.

We also need the 3D equation of state with columns (n_B, Y_e, T) . This table will be used to calculate the specific enthalpy. A step-by-step approach is:

1. Take the points for the density, electron fraction, and the temperature from the `sph` output and clip the points to the ranges of the 3D table. Change the density and temperature columns to log-scale.
2. Set up interpolators to interpolate the total energy density $\frac{e}{n_B m_n}$ and the pressure $\frac{p}{n_B m_n}$ from the table given $(\log_{10}(n_B), Y_e, \log_{10}(T))$.
3. For all the points on the spherical shell, interpolate to obtain the total energy density and the pressure and then compute the specific enthalpy via

$$h = 1 + e_{\text{intp}} + p_{\text{intp}}.$$

4. Compute the Lorentz-factor via Equ. (??).
5. Compute the 4-velocity with upper indices u^i via Eqs. (??)-(??).
6. Compute the radial component from the spatial part of the 4-velocity with Equ. (??).
7. Lower the index on the time-component of the 4-velocity u_t via Equ. (??).
8. Create masks, such that the density is above the density floor and that the radial component of the velocity u^r is greater than 0.
9. Compute a criterion mask with

$$\begin{aligned} \text{Bernoulli : } & h \cdot u_t < h_{\min}, \\ \text{Geodesic : } & u_t < -1. \end{aligned}$$

10. Compute the determinant of the spatial metric γ with Equ. (??) and the base integrand via

$$I = \rho u^r \sqrt{\gamma} r^2 \sin \theta.$$

11. See the documentation on the ejecta and neutrino analysis in the repo for more information about the various quantities one can calculate with the above steps.

NEUTRINO LUMINOSITY AND AVERAGE ENERGY DENSITY For the neutrino luminosity and average energy density we again require spherical output at a fixed radius, i.e.


```
<output4>
file_type = sph
variable = rad_m1_F
dt = 10.0
radius = 400.0
ntheta = 256
```

The variables we need are:

- The ADM metric components `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, and `adm_gzz` or the entire ADM output block `adm`. Note: if the entire `adm` block is specified, then the six extrinsic curvature components are also stored which are not needed for this computation and might fill unnecessary disk space.
- The gauge variables `z4c_alpha`, `z4c_betax`, `z4c_betay`, and `z4c_betaz`.
- The neutrino energy density per species `rad_m1_E`, the neutrino number flux per species `rad_m1_F`, and the neutrino number density per species `rad_m1_N` in the Eulerian frame.

A step-by-step guide is:

1. Take the neutrino number flux F_i and raise its components with the inverse spatial metric, which can be computed with Eqs. (??)-(??). The raising operation is then defined via Eqs. (??)-(??). We obtain the vector F^i .
2. Taking the raised neutrino number flux vector in the Eulerian frame F^i , we can compute the metric-normalized radial projection F_{norm}^r via Equ. (??).
3. The same projection is done for the shift vector β^i to obtain β_{norm}^r .
4. The radial flux is compute as

$$F_{\text{radial}} = \alpha F^r - \beta^r E.$$

5. Compute the neutrino luminosity per species s as

$$L_s^{\text{code}} = \sum F_{\text{radial}} w,$$

where w are the surface weights of the extraction sphere. The luminosity has to be converted into physical units afterwards.

6. Compute the average energy density of the neutrino species, by computing the number flux as

$$N_{\text{flux}} = \frac{F_{\text{radial}} N_{\text{code}}}{E},$$

where N_{code} is the neutrino number density in code units.

7. Lastly, compute the change of the neutrino number density as

$$\dot{N}_{\text{code}} = \sum N_{\text{flux}} w,$$

and then compute the average energy density in code units as

$$\langle \varepsilon \rangle_s = \frac{L_s^{\text{code}}}{\dot{N}_{\text{code}}}.$$

Note: the exact M1 number flux uses f_ν^a , so the above is an approximation.

■ M1 VARIABLES IN FLUID FRAME Since all the M1 variables in AthenaK are given in the Eulerian frame, we need to perform a set of projections to obtain them in the fluid frame. This is particularly desirable if we want to do a tracer analysis in post-processing. To convert the M1 variables, namely the radiation energy density E , the radiation flux F^α ,

and the radiation pressure tensor $P^{\alpha\beta}$ in the Eulerian frame into the fluid frame, we first note that the definition of the radiation stress-energy tensor in both frames is

$$\text{Eulerian frame : } T^{\alpha\beta} = En^\alpha n^\beta + F^\alpha n^\beta + n^\alpha F^\beta + P^{\alpha\beta}, \quad (41)$$

$$\text{Fluid frame : } T^{\alpha\beta} = Ju^\alpha u^\beta + H^\alpha u^\beta + u^\alpha H^\beta + K^{\alpha\beta}, \quad (42)$$

where n^α is the unit-normal to the $t = \text{const.}$ hypersurface, u^α is the fluid four-velocity, J is the radiation energy density, H^α the radiation flux, and $K^{\alpha\beta}$ the radiation pressure tensor in the fluid frame. We will need to carry out a set of projections to obtain the fluid frame variables. An additional quantity in the fluid frame will be the comoving number density $n = \frac{N}{\Gamma}$, where N is the number density in the Eulerian frame and Γ the ratio of neutrinos in the Eulerian and the fluid frame. Since this is not an evolved variable, we can simply derive it on the way from the evolved ones. Note: the procedure has to be done per species. A step-by-step approach is:

1. Compute the lower and upper spacetime metric from the spatial metric γ_{ij} and the gauge variables α , and β^i via

$$\begin{aligned} g_{00} &= -\alpha^2 + \beta^i \beta_i, \\ g_{0j} &= \beta_j, \\ g_{ij} &= \gamma_{ij}, \\ g^{00} &= -\frac{1}{\alpha^2}, \\ g^{0j} &= \frac{\beta^j}{\alpha^2}, \\ g^{ij} &= \gamma^{ij} - \frac{\beta^i \beta^j}{\alpha^2}. \end{aligned}$$

These can be computed via the functions `adm::SpacetimeMetric` and `adm::SpacetimeUpperMetric` in `adm.hpp`.

2. Populate a four-vector n_α using `pack_n_d` inside of `radiation_m1_tensors.hpp`, which defines

$$\begin{aligned} n_0 &= -\alpha, \\ n_i &= 0. \end{aligned} \quad (43)$$

3. Populate a four-vector β^μ with the spatial components of the shift β^i , where for $\mu = 0$ the vector is zero (via the function `pack_beta_u` in `radiation_m1_tensors.hpp`).
4. Compute the Lorentz-factor using Equ. (??), and the Lorentz-weighted 3-velocity in AthenaK by using the function `get_w_lorentz` inside `radiation_m1_tensors.hpp`.
5. Pack a vector for the 4-velocity u^α via `pack_u_u` with

$$\begin{aligned} u^0 &= \frac{W}{\alpha}, \\ u^i &= \tilde{u}^i - \frac{W\beta^i}{\alpha}, \end{aligned}$$

and pack a vector for the 3-velocity v^α (only spatial components important) via `pack_v_u` with

$$\begin{aligned} v^0 &= 0, \\ v^i &= \frac{u^i}{W} + \frac{\beta^i}{\alpha}. \end{aligned}$$

Above, $\tilde{u}^i$ is the Lorentz-weighted 3-velocity that AthenaK uses.

6. Use the `tensor_contract` function in `radiation_m1_tensors.hpp` to compute u_α and v_α using the lower spacetime metric $g_{\alpha\beta}$ from step 1.
7. Calculate the fluid projector P_β^α from u_α and u^α using `calc_proj` inside `radiation_m1_helper.hpp` which computes

$$P_\beta^\alpha = \delta_\beta^\alpha + u^\alpha u_\beta,$$

where δ_β^α is the Kronecker-delta.

8. For each neutrino species, get the radiation energy density E from the array holding the solution and pack the radiation flux F^α using `pack_F_d` in `radiation_m1_tensors.hpp`, which computes

$$F_0 = \beta^i F_i,$$

$$F_i = F_i,$$

where F_i is the radiation flux per species evolved on the given hypersurface.

9. Using the spacetime metric with lower indices $g_{\alpha\beta}$, with upper indices $g^{\alpha\beta}$, the unit-normal with lower index n_α , the Lorentz-factor W , the fluid 4-velocity u^α , the fluid 3-velocity v_i , the fluid projector P_β^α , the radiation energy density E , and the radiation flux F_α in the Eulerian frame plus the Eddington-factor χ , we can compute the radiation stress-energy tensor $P_{\alpha\beta}$ using the `apply_closure` function inside `radiation_m1_helpers.hpp`. This function computes

$$P_{\alpha\beta}^{\text{thin}} = \frac{E}{F_\mu F^\mu} F_\alpha F_\beta,$$

$$\gamma_\alpha^\kappa H_\kappa = \frac{F_\alpha}{W} + \frac{W}{2W^2 + 1} v_\alpha ((4W^2 + 1)F^\mu v_\mu - 4W^2 E),$$

$$P_{\alpha\beta}^{\text{thick}} = \left(\frac{2W^2 - 1}{2W^2 + 1} E - 2W^2 F^\mu v_\mu \right) (4W^2 v_\alpha v_\beta + g_{\alpha\beta} + n_\alpha n_\beta)$$

$$+ W(\gamma_\alpha^\kappa H_\kappa v_\beta + \gamma_\beta^\kappa H_\kappa v_\alpha),$$

$$P_{\alpha\beta} = \frac{3\chi - 1}{2} P_{\alpha\beta}^{\text{thin}} + \frac{3(1 - \chi)}{2} P_{\alpha\beta}^{\text{thick}}.$$

10. Using the radiation pressure tensor in the Eulerian frame $P_{\alpha\beta}$, we can now compute the stress-energy tensor $T_{\alpha\beta}$ in the Eulerian frame using the function `assemble_rT` in `radiation_m1_helpers.hpp`, which computes

$$T_{\alpha\beta} = E n_\alpha n_\beta + F_\alpha n_\beta + n_\alpha F_\beta + P_{\alpha\beta}.$$

11. With the stress-energy tensor in the Eulerian frame $T_{\alpha\beta}$, we can now contract this with the four-velocity vector to obtain the radiation energy density in the fluid frame via

$$J = T_{\alpha\beta} u^\alpha u^\beta. \quad (44)$$

This computation is encapsulated inside the function `calc_J_from_rT` inside `radiation_m1_helpers.hpp`.

12. Using the fluid projector P_β^α , the stress-energy tensor in the Eulerian frame $T_{\alpha\beta}$ and the fluid four-velocity u^α , we can compute the radiation flux in the fluid frame H_α via

$$H_\alpha = -P_\alpha^\beta u^\kappa T_{\beta\kappa}. \quad (45)$$

This computation is encapsulated inside the function `calc_H_from_rT` inside `radiation_m1_helpers.hpp`. Contract with the spacetime metric $g^{\alpha\beta}$ to obtain H^α .

13. Enforce that J and H_α are above their respective floors with `apply_floor` inside `radiation_m1_helpers.hpp`.
14. Compute the comoving number density n , using J , E , and F_α . For that we compute the ratio of neutrinos in the Eulerian and the fluid frame Γ via `compute_Gamma` inside `radiation_m1_helpers.hpp` which defines

$$\Gamma = W \frac{E}{J} \left(1 - \frac{F_\alpha v^\alpha}{E} \right), \quad n = \frac{N}{\Gamma}, \quad (46)$$

where N is the number density in the Eulerian frame, and v^α the 3-velocity packed into a 4-vector.

15. Optional: compute the unit-norm radiation number current f_ν using the fluid four-velocity u^α , the radiation energy density J , and the radiation flux in the fluid frame H^α with

$$f_\nu^\alpha = u^\alpha + \frac{H^\alpha}{J}. \quad (47)$$

The computation is inside the function `assemble_fnu` in `radiation_m1_helpers.hpp`. Note: inside the computation there is a check that the radiation energy density in the fluid frame J is above the floor for the radiation energy density in the Eulerian frame E . If this is not the case, the second term vanishes.